Class 3

Relativistic Heavy Ions & HEP Jets

Bridging QGP Phenomenology and Collider Jet Physics

Duration: 3 months · 13 weeks · ~5–10 hrs/week
Part of the 2026–2027 Self-Study Plan · RHI → HEP Transition

Uses your existing RHI expertise as a launch pad into HEP jet physics. Jet quenching is the natural bridge. Three months of doing both at once makes both easier.

Course Overview

This class is purpose-built for the RHI → HEP transition. You already know heavy-ion physics professionally; the goal here is two-fold. First, to shore up your self-taught foundations in QCD at high energy — factorization, PDFs, evolution, Monte Carlo — so that they stand up to HEP-theorist scrutiny. Second, to build genuine fluency in HEP jet physics: algorithms, substructure, grooming, tagging, and the modern ML methods that now dominate the field.

Jet quenching is the natural bridge. A boosted-object analysis in pp and a jet-R_{AA} measurement in PbPb use almost the same theoretical and computational machinery; the hardware you already understand is the same hardware ATLAS and CMS use. What differs is the emphasis: in HEP you care about substructure-level tagging of boosted W/Z/H/top; in heavy ions you care about medium modification of that same substructure.

By the end you should be able to contribute to a HEP jet analysis and to articulate, to a HEP audience, what your heavy-ion experience buys you.

Learning Goals

- Work confidently with collinear factorization, DGLAP evolution, and modern PDF sets.
- Understand jet algorithms (anti-k_T, k_T, C/A, SISCone) well enough to implement one in Python.
- Run Pythia 8 and FastJet from a Jupyter notebook; produce your own jet observables from simulated events.
- Explain grooming (mMDT, Soft Drop), N-subjettiness, and energy correlation functions to a seminar audience.
- Read jet-quenching theory (Mehtar-Tani, Wiedemann, Salgado, Vitev) with comprehension.
- Articulate how HEP jet substructure methods are being adapted for QGP tomography.

Prerequisites & Background Check

Mostly stuff you already have from RHI. The gap is probably on the HEP-specific jet tooling and ML side.

- RHI phenomenology at the level you already practice.
- QCD Lagrangian at the working level of QFT II (or assume now, revisit in parallel).
- Comfort with ROOT and/or Python + NumPy + uproot.
- Basic understanding of collider kinematics: rapidity, p_T, pseudorapidity, luminosity.

Textbooks & Core Resources

Primary Textbooks

- **Marzani, Soyez & Spannowsky**, *Looking Inside Jets* — the definitive jet substructure introduction. The spine of the second half.
- **Ellis, Stirling & Webber**, *QCD and Collider Physics* (the Pink Book) — still the best collider-QCD foundation.
- **Florkowski**, *Phenomenology of Ultra-Relativistic Heavy-Ion Collisions* — modern HI reference; skim familiar chapters.

Supplementary / Reference

- **Campbell, Huston & Krauss**, *The Black Book of QCD* — encyclopedic modern QCD.
- **Wong**, *Introduction to High-Energy Heavy-Ion Collisions*.
- **Yagi, Hatsuda & Miake**, *Quark-Gluon Plasma*.
- **Kovchegov & Levin**, *QCD at High Energy* — for small-x and saturation.
- **Collins**, *Foundations of Perturbative QCD* — for factorization in rigor.

Online Lectures & Notes

- **FastJet manual** (fastjet.fr) — read and use.
- **Pythia 8 manual and worksheet** (home.thep.lu.se/Pythia/) — run the full worksheet in Week 4.
- TASI lectures on jets: Salam 2010, Larkoski 2017, Nachman 2020.
- BOOST workshop proceedings (annual) — current tagging state of the art.
- Mehtar-Tani, Milhano & Tywoniuk, *Jet Quenching: a Primer* (arXiv:1302.2579).
- Quark Matter, Hard Probes, Initial Stages proceedings — stay current.

Key Papers & Reviews

- Cacciari, Salam & Soyez, *The anti-k_T jet clustering algorithm* (2008).
- Salam, *Towards Jetography* (2010).
- Butterworth, Davison, Rubin & Salam, *Jet substructure as a new Higgs search channel at the LHC* (2008).
- Larkoski, Marzani, Soyez & Thaler, *Soft Drop* (2014).
- Thaler & Van Tilburg, *N-subjettiness* (2011).
- Qin & Wang review on jet quenching (Int. J. Mod. Phys. E).
- Cao & Wang review of jet modifications in the QGP.
- ATLAS/CMS jet-R_{AA} and substructure-in-PbPb papers.

13-Week Schedule

Heavy emphasis in Weeks 5–10 on actually running code. If you have not used Pythia + FastJet before, give yourself extra time in Week 4.

Week	Topic	Readings / Activities	Deliverable
1	Collider QCD foundations	ESW Chs. 1-3; PDG QCD review	Notes on factorization
2	PDFs & DGLAP evolution	ESW Ch. 4; modern LHAPDF docs	Plot xf_i(x,Q<super>2</supe
3	Jet algorithms	Marzani-Soyez-Spannowsky Ch. 2; FastJet manual	Python toy anti-k_T
4	Parton showers; Pythia 8	Pythia 8 worksheet; Sjöstrand reviews	Run Pythia, extract p_T spec
5	Jet cross sections, NLO basics	ESW Ch. 5; MCFM or MadGraph demo	Reproduce an inclusive jet plot
6	Jet substructure I: grooming	Marzani-Soyez-Spannowsky Chs. 3-4	mMDT + Soft Drop in your Pythia events
7	Jet substructure II: N-subj, ECFs	Marzani-Soyez-Spannowsky Chs. 5-6; Thaler-Van Tilburg	Plot τ₂₁ for QCD vs W
8	Boosted tagging: W, Z, H, top	Marzani-Soyez-Spannowsky Ch. 7; BOOST proceedings	Construct a simple boosted-W tagger
9	ML for jets	Larkoski-Moult-Nachman review; jet-images paper (de Oliveira et al.)	Train a small CNN on jet images
10	Jets in heavy ions I: quenching theory	Mehtar-Tani et al. primer; BDMPS-Z review	Derive E-loss parametrics
11	Jets in heavy ions II: observables	ATLAS/CMS jet-R_{AA} and groomed-mass-in-PbPb papers	Reproduce jet-R_{AA} plot
12	Small systems, saturation, EIC	Kovchegov-Levin Ch. 4-5; EIC Yellow Report (excerpts)	Short note on saturation signatures
13	Synthesis; capstone	Re-read Marzani-Soyez-Spannowsky Ch. 8	Capstone writeup

Pacing assumes ~5–10 hrs/week on this course while running in parallel with one other class. If time is tighter, cut problem sets first; if time is looser, add problems and deeper paper reading.

Problem Sets & Exercises

This class is heavier on coding than on textbook problems. For theory problems, use ESW end-of-chapter problems and the exercises in Marzani-Soyez-Spannowsky.

- Implement anti-k_T in Python from scratch for ~50 particles; compare output to FastJet on the same event.
- Run Pythia 8 pp → dijet at $\sqrt{s}$ = 13 TeV; extract and plot the inclusive jet p_T spectrum. Compare shape to ATLAS/CMS public data.
- Generate W→qq' and QCD dijet samples; plot groomed jet mass for both. Quantify signal-background separation.
- ESW end-of-chapter problems on DGLAP splitting functions and k_T-factorization.
- Marzani-Soyez-Spannowsky exercises in Chs. 3–7.
- Modify your Pythia workflow to dump to HepMC; read with uproot; build an analysis pipeline you could reuse for another study.
- Train a simple graph neural network (one PyTorch Geometric tutorial modification) on top-vs-QCD jets from the public top-tagging dataset.

Capstone Project

Two deliverables: one code-heavy, one writing-heavy, turned in together as a portfolio piece you could show a HEP collaboration.

- **Code piece.** Produce a publication-quality plot that reproduces, qualitatively, one published result you care about — for example, the ATLAS or CMS boosted-W tagger ROC curve, or a groomed-jet mass distribution in PbPb vs pp. Pipeline: Pythia→FastJet→your analysis code→Matplotlib. Include the full code in a Git repo.
- **Writing piece.** ~8 page whitepaper: *Leveraging Heavy-Ion Expertise for Jet Physics in pp*. Articulate explicitly what transfers from your RHI work (medium effects, flow, multiplicity) and what is new (substructure tagging, boosted kinematics, pileup). This doubles as networking material.

Milestones & Self-Assessment

At each checkpoint, you should be able to do the listed tasks without looking them up. If you can't, go back rather than pushing forward.

- **End of Week 4:** You have a working Pythia + FastJet analysis pipeline in Python, running on your laptop.
- **End of Week 7:** You can explain soft-drop grooming and N-subjettiness at a board, without notes.
- **End of Week 10:** You can describe BDMPS-Z radiative energy loss at the parametric level and identify which observables it predicts.
- **End of Week 13:** You can walk into a HEP jet-substructure seminar and follow 80%+ of the talk, and you have a portfolio piece you can point to.

End of class plan. Iterate after the first two weeks — what was over-ambitious gets cut; what felt thin gets deepened.